

Shift-Aware Reliability Enhancement for Machine Learning-Based NIDS Under Cross-Dataset Distribution Shift

Khondokar Sajid (2211954042) Kazi Safin Arafat (2211778642)
Moushumi Akter Mow (2021983642) Rakibul Hasan Ridoy (1731339042)

Department of Electrical and Computer Engineering

North South University, Dhaka, Bangladesh

Faculty Advisor: Md. Mohammad Shifat-E-Rabbi

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Abstract—ML-based Network Intrusion Detection Systems (NIDS) often achieve high benchmark accuracy but suffer silent reliability degradation under cross-dataset distribution shift. This proposal addresses deployment realism by (i) evaluating how shift degrades confidence calibration using Expected Calibration Error (ECE) and Brier Score, (ii) applying post-hoc recalibration (Temperature Scaling and Isotonic Regression), and (iii) introducing an entropy-based Out-of-Distribution (OOD) rejection gate that outputs *UNKNOWN* when predictions are unreliable. The result is a deployment-aware NIDS suited for heterogeneous environments including IoT-heavy networks.

Index Terms—Network Intrusion Detection, Distribution Shift, Reliability, Probability Calibration, Expected Calibration Error, OOD Detection, IoT Security

I. INTRODUCTION

ML-based NIDS learn discriminative patterns from labeled traffic data and can outperform signature-based approaches on benchmarks. However, a critical assumption—that training and deployment traffic share the same distribution—is routinely violated in practice. Device ecosystems, application mixes, and policies vary across sites, causing **cross-dataset distribution shift** that can silently erode both detection performance and the reliability of confidence scores. Miscalibration under shift is particularly dangerous: alert prioritization and automated response decisions both rely on confidence scores being trustworthy. An

overconfident model may assign high certainty to incorrect decisions, producing missed attacks or persistent false-alarm fatigue.

In CSE499A, we built and validated a functional ML-based NIDS under standard same-dataset protocols. In CSE499B, we ask: *Can the model recognize and communicate its own uncertainty under distribution shift?* We design a lightweight post-hoc reliability framework that quantifies, reduces, and communicates overconfidence under cross-domain deployment.

Problem Definition

Core problem. An ML-based NIDS trained on Domain A and deployed on Domain B may undergo **calibration collapse**: confidence scores no longer reflect true correctness probability. Even when accuracy partially holds, the model may be systematically overconfident on misclassified samples—a failure mode invisible to Accuracy and F1 metrics.

Objective. (1) Quantify reliability degradation via ECE, Brier Score, and reliability diagrams. (2) Reduce unsafe overconfidence via deployable post-hoc mechanisms requiring no retraining.

Motivation and Practical Relevance

Practical relevance. Networks in Bangladesh commonly use low-cost routers and heterogeneous IoT devices (IP cameras, smart TVs, biometric systems). Background traffic in these

environments differs markedly from controlled lab datasets. Under such shift, an overconfident NIDS may miss malicious activity with high certainty (false negatives) or flood operators with false alarms (alert fatigue). Calibrated confidence outputs and an explicit *UNKNOWN* decision class let operators focus on reliable alerts and escalate uncertain cases for human review.

Proposed Contribution

Contributions. We introduce a **deployment reliability framework** applicable to any trained binary NIDS classifier:

- 1) **Cross-dataset reliability evaluation:** Calibration benchmarking in the $A \rightarrow B$ transfer setting using ECE, Brier Score, and reliability diagrams.
- 2) **Post-hoc probability calibration:** Temperature Scaling and Isotonic Regression to recover trustworthy confidence estimates without retraining.
- 3) **Entropy-based OOD rejection:** Suppresses overconfident predictions by returning *UNKNOWN* when predictive entropy exceeds a learned threshold.

II. RELATED WORK

Guo et al. [1] showed that deep classifiers are often overconfident and that temperature scaling substantially improves calibration. Zadrozny and Elkan [2] and Niculescu-Mizil and Caruana [3] established calibration foundations for tree-based models. Ovadia et al. [5] demonstrated that predictive uncertainty degrades under dataset shift, while Hendrycks and Gimpel [6] proposed maximum softmax probability as a baseline OOD detector. Selective prediction—allowing models to abstain on low-confidence inputs—was formalized by Geifman and El-Yaniv [8]. In NIDS literature, most studies evaluate on same-distribution splits of UNSW-NB15 [9], NSL-KDD [10], and TON_IoT [11], leaving cross-dataset reliability largely unaddressed.

III. PROPOSED FRAMEWORK

Figure 1 illustrates the pipeline. Our framework appends a calibration stage and an entropy-based rejection gate to any trained classifier, without modifying it.

IV. METHODOLOGY AND EXPERIMENTAL DESIGN

A. Datasets

- **TON_IoT** [11]: IoT-focused; primary training environment.
- **UNSW-NB15** [9] or **NSL-KDD** [10]: General-purpose benchmarks used as the deployment-target environment.

A **binary task** (Benign vs. Attack) ensures stable cross-dataset label correspondence.

B. Feature Alignment

A **universal flow-level feature set** (15–25 features) is derived from fields available across both datasets: flow duration, byte/packet counts, inter-arrival time, bytes/sec, packets/sec, and protocol indicators. All datasets are projected into this common space before training or evaluation.

C. Classifiers

Three architectures covering distinct calibration behaviors: **Random Forest** (stable ensemble baseline), **XGBoost** (high-performance, prone to overconfidence under shift), and **MLP** (lightweight DNN, known to be poorly calibrated [1]).

D. Evaluation Metrics

Discriminative: Accuracy, Precision, Recall, F1-score, ROC-AUC.

Reliability: ECE, Brier Score, and reliability diagrams.

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (1)$$

$$\text{BS} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2 \quad (2)$$

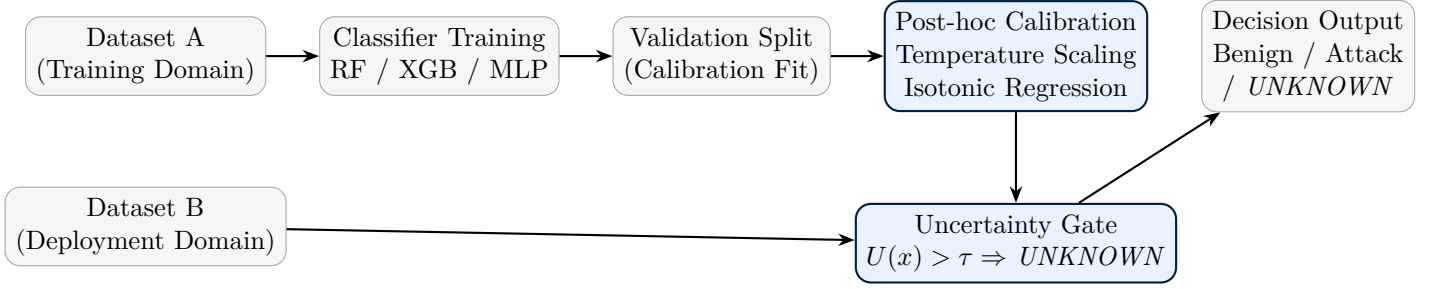


Fig. 1. Deployment reliability framework: a post-hoc calibration module and entropy-based OOD gate are appended to the trained classifier for cross-domain deployment on Dataset B.

Algorithm 1 Shift-Aware Reliability Enhancement Protocol

Require: Datasets A, B ; feature set $\mathcal{F}$; classifiers $\mathcal{M}$

- 1: Project A, B into $\mathcal{F}$; partition A into $A_{\text{train}}/A_{\text{val}}/A_{\text{test}}$ (70/15/15, stratified)
- 2: **for** each classifier $m \in \mathcal{M}$ **do**
- 3: Train m on A_{train}
- 4: Evaluate on A_{test} (Acc/F1/AUC + ECE/Brier) $\triangleright$ in-distribution baseline
- 5: Evaluate on B_{test} (same metrics + reliability diagram) $\triangleright$ cross-dataset baseline
- 6: Fit calibrator on A_{val} ; re-evaluate ECE/Brier on B_{test}
- 7: Determine τ from A_{val} ; apply OOD gate on B_{test}
- 8: Report Coverage, Selective Accuracy, and $UNKNOWN$ rate
- 9: **end for**

E. Calibration and OOD Gate

Temperature Scaling [1] (single parameter; effective for neural models) and **Isotonic Regression** [2] (non-parametric; effective for tree-based models) are fitted on A_{val} and evaluated on B_{test} . Predictive entropy $U(x) = -\sum_{i=1}^C p_i \log p_i$ is computed on calibrated outputs. Samples with $U(x) > \tau$ are assigned $UNKNOWN$; the threshold τ is selected on A_{val} and applied unchanged on B_{test} [8], [6].

TABLE I
EXPERIMENTAL DELIVERABLE CHECKLIST

Deliverable	Description
Cross-dataset tables	Discriminative + calibration metrics for $A \rightarrow B$ transfer
Reliability diagrams	Confidence vs. accuracy plots before and after recalibration
Calibration summary	ECE and Brier Score comparison per model (uncalibrated vs. calibrated)
OOD analysis	Coverage vs. Selective Accuracy trade-off curves
Attack breakdown	Per-category error and uncertainty analysis

F. Experimental Protocol

V. PLANNED OUTPUTS AND ANTICIPATED FINDINGS

Three primary findings are anticipated. **(F1)** Cross-dataset deployment produces measurably higher ECE and Brier Scores than in-distribution evaluation, confirming calibration collapse even when discriminative metrics partially hold. **(F2)** Post-hoc calibration reduces ECE and Brier Score on B_{test} , demonstrating recoverability without retraining. **(F3)** The entropy gate identifies uncertain inputs and assigns $UNKNOWN$, yielding meaningfully higher Selective Accuracy on accepted samples.

VI. CSE499A vs. CSE499B SCOPE

VII. CONCLUSION

This proposal argues that accuracy alone is insufficient for evaluating NIDS intended for het-

TABLE II
SCOPE COMPARISON: CSE499A AND CSE499B

Dimension	CSE499A	CSE499B
Objective	Accuracy + functional NIDS	Reliability + safe deployment
Evaluation	Same-dataset splits	Cross-dataset transfer
Confidence	Not examined	ECE, Brier, reliability diagrams
Output	Binary label	Binary + UNKNOWN
Novelty	Functional baseline	Reliability framework

erogeneous real-world deployment. Cross-dataset distribution shift—an unavoidable feature of real network environments—can silently degrade confidence reliability, transforming a nominally accurate classifier into a system whose outputs cannot be safely trusted by operators.

The proposed three-component framework addresses this gap: a cross-dataset calibration evaluation pipeline, post-hoc recalibration via Temperature Scaling and Isotonic Regression, and an entropy-based OOD gate enabling selective abstention. All components are lightweight, retraining-free, and compatible with any binary classifier, making the framework accessible for practical deployment in resource-constrained settings such as IoT gateways and small office environments. The work also contributes evidence that calibration-aware evaluation should become a standard component of NIDS benchmarking alongside conventional discriminative metrics.

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REFERENCES

- [1] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. ICML*, 2017.
- [2] B. Zadrozny and C. Elkan, “Obtaining calibrated probability estimates from decision trees and naive Bayesian classifiers,” in *Proc. ICML*, 2001.
- [3] A. Niculescu-Mizil and R. Caruana, “Predicting good probabilities with supervised learning,” in *Proc. ICML*, 2005.
- [4] J. Platt, “Probabilistic outputs for support vector machines,” in *Advances in Large Margin Classifiers*, MIT Press, 1999.
- [5] Y. Ovadia *et al.*, “Can you trust your model’s uncertainty? Evaluating predictive uncertainty under dataset shift,” in *NeurIPS*, 2019.
- [6] D. Hendrycks and K. Gimpel, “A baseline for detecting misclassified and out-of-distribution examples in neural networks,” in *ICLR*, 2017.
- [7] F. T. Liu, K. M. Ting, and Z.-H. Zhou, “Isolation forest,” in *Proc. ICDM*, 2008.
- [8] Y. Geifman and R. El-Yaniv, “Selective prediction in deep neural networks,” in *Proc. NeurIPS*, 2017.
- [9] N. Moustafa and J. Slay, “UNSW-NB15: A comprehensive data set for network intrusion detection systems,” in *Proc. MilCIS*, 2015.
- [10] M. Tavallaei, E. Bagheri, W. Lu, and A. A. Ghorbani, “A detailed analysis of the KDD CUP 99 data set,” in *Proc. IEEE CISDA*, 2009.
- [11] A. Alsaedi, N. Moustafa, Z. Tari, A. Anwar, and A. Mahmood, “TON_IoT telemetry dataset,” *IEEE Internet of Things J.*, vol. 7, no. 9, pp. 7879–7886, 2020.
- [12] B. Lakshminarayanan, A. Pritzel, and C. Blundell, “Simple and scalable predictive uncertainty estimation using deep ensembles,” in *Proc. NeurIPS*, 2017.